

1. (Stewart, Chapter 5 Review, #39 and #92)

(a) Show that

$$\cos(\arctan(\sin(\operatorname{arccot} x))) = \sqrt{\frac{x^2 + 1}{x^2 + 2}}$$

(b) Differentiate $y = \ln(\cosh 3x)$.

2. (Stewart, Chapter 5 Review, #27 and #88)

(a) Evaluate the integral $\int \frac{x}{\sqrt{1-x^4}} dx$.

(b) Differentiate $y = x \sinh(x^2)$

3. (Stewart, Chapter 5 Review, #41 and #82)

(a) Evaluate the integral $\int_0^{\pi/2} \frac{\cos x}{1 + \sin^2 x} dx$.

(b) Differentiate $y = \cosh^{-1}(\sinh x)$

4. (Stewart, Chapter 5 Review, #64, and §5.7, #61)

(a) Evaluate the limit $\lim_{x \rightarrow \infty} \arctan(x^3 - x)$.

(b) Evaluate the integral $\int \frac{e^x}{1 - e^{2x}} dx$.

5. (Stewart, Chapter 5 Review, #81, and §5.7, #43)

(a) Differentiate $y = \arctan(\arcsin \sqrt{x})$.

(b) Show that $\frac{d}{dx} \arctan(\tanh x) = \operatorname{sech} 2x$